Nama: Muhammad Bintang Gunawan

1. Instruksi Tugas

Mengimplementasikan fitur CRUD (Create, Read, Update, Delete) pada tabel Transaksi menggunakan framework Laravel.

2. Implementasi Program

2.1 Model Transaction

Model Transaction saya buat di file:
app/Models/Transaction.php

Di model ini saya menentukan tabelnya, yaitu transactions, dan mengatur kolom yang boleh diisi lewat $fillable. Saya juga membuat dua relasi:

- belongsTo(User::class, 'customer_id') untuk menghubungkan transaksi dengan data pengguna.
- belongsTo(Book::class) untuk menghubungkan transaksi dengan data buku.

Dua relasi ini dipakai untuk mengambil informasi pengguna dan buku yang terkait dengan sebuah transaksi.

2.2 Migration Tabel Transactions

Saya membuat migration untuk menentukan struktur tabel transaksi. Kolom yang dipakai:

- order_number: nomor pesanan yang unik.
- customer_id: foreign key yang mengarah ke tabel users.
- book_id: foreign key yang mengarah ke tabel books.
- quantity: jumlah buku yang dibeli.
- total_amount: total harga transaksi.

Foreign key dibuat dengan foreignId dan memakai aturan cascade delete.

2.3 TransactionController

Controller ini saya buat di file:
app/Http/Controllers/TransactionController.php

Isinya ada lima fungsi utama:

- Store: membuat transaksi baru, menghitung total harga dari harga buku dikali jumlah pembelian, lalu mengurangi stok buku.
- Index: mengambil semua data transaksi beserta relasi pengguna dan bukunya.
- Show: menampilkan detail transaksi berdasarkan ID.
- Update: memperbarui data transaksi dan menyesuaikan stok buku.
- Destroy: menghapus transaksi dan mengembalikan stok buku.

2.4 Middleware Customer dan Admin

Untuk mengatur hak akses, saya memakai dua middleware.

AdminMiddleware: hanya mengizinkan pengguna dengan role admin untuk mengakses fitur Read All dan Destroy.

CustomerMiddleware: hanya mengizinkan pengguna dengan role user untuk mengakses fitur Create, Update, dan Show.

Keduanya memakai autentikasi Laravel Sanctum untuk memastikan pengguna sudah login.

2.5 Pengaturan Routing

Routing saya atur di file:
routes/api.php

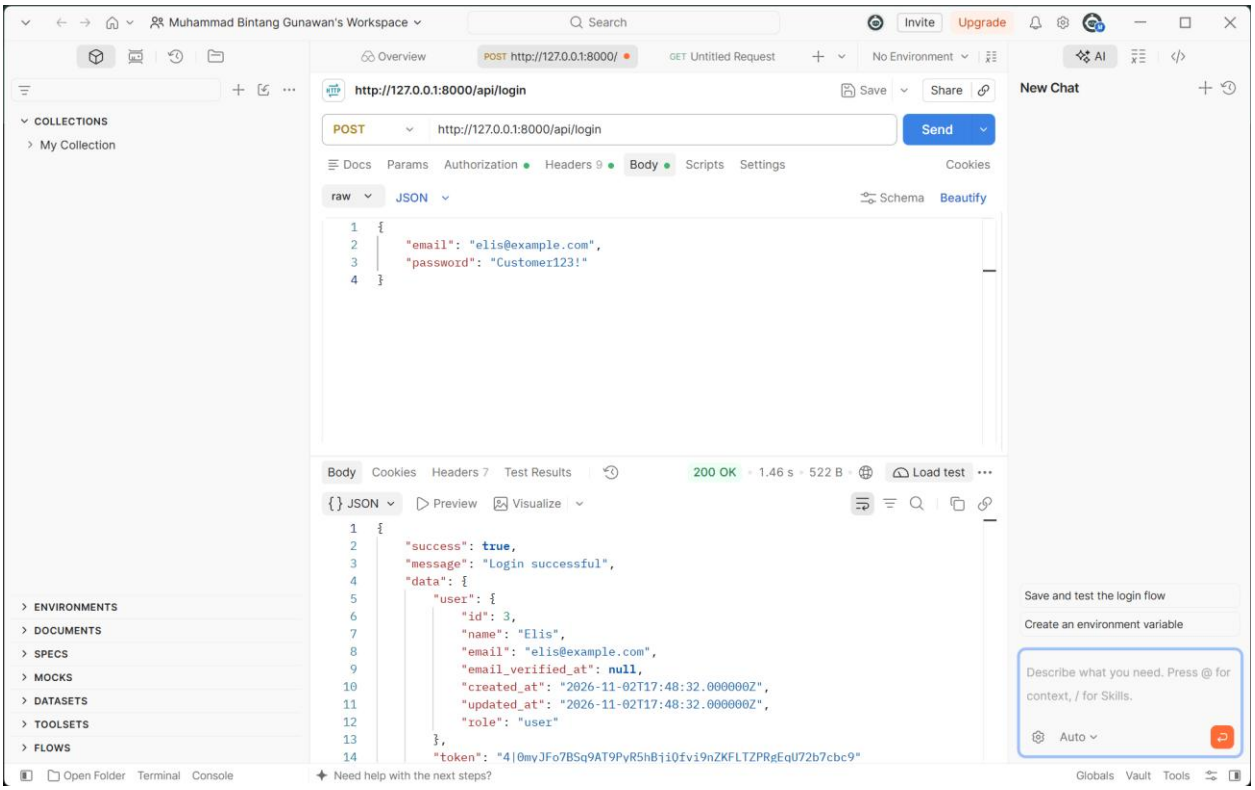
Pembagian endpoint berdasarkan hak aksesnya:

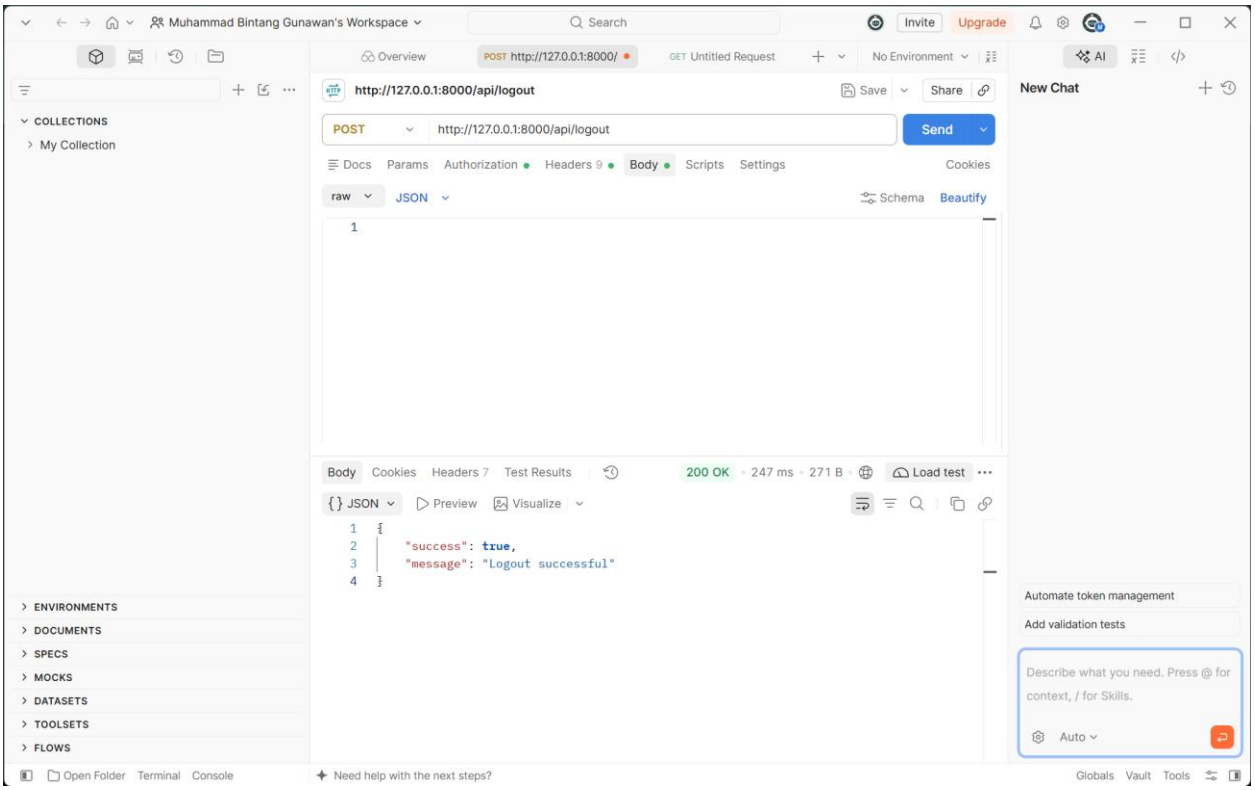
- POST /api/transactions: hanya untuk Customer.
- GET /api/transactions/{id}: hanya untuk Customer.
- PUT /api/transactions/{id}: hanya untuk Customer.
- GET /api/transactions: hanya untuk Admin.
- DELETE /api/transactions/{id}: hanya untuk Admin.

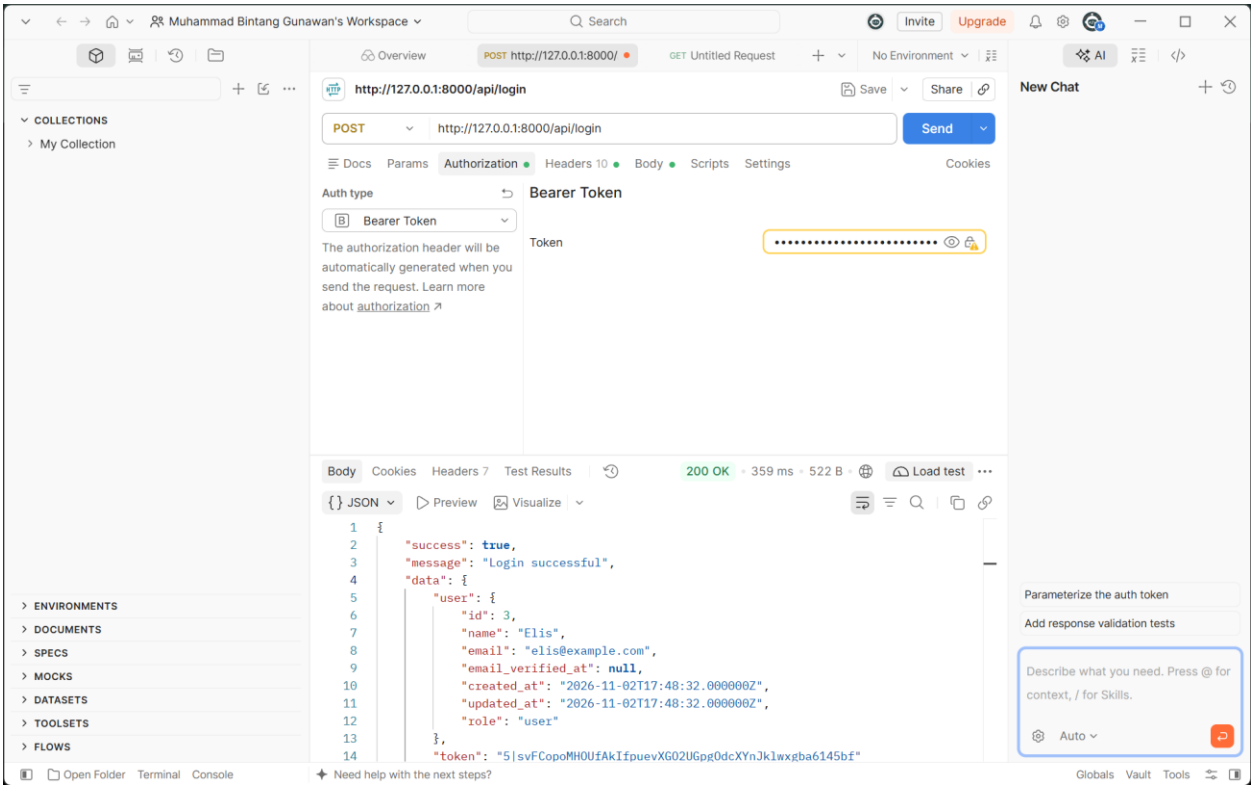
2.6 Autentikasi API

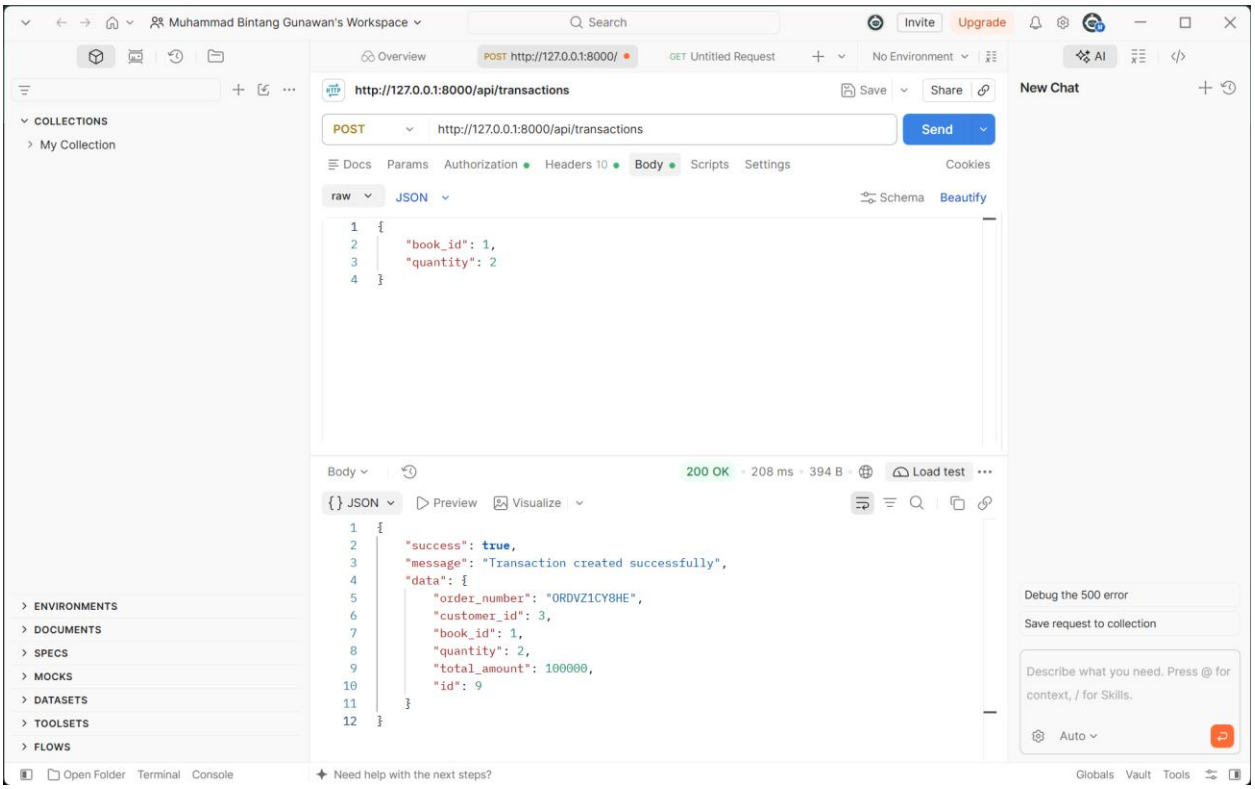
Saya membuat AuthController untuk menangani login dan logout dengan Laravel Sanctum. Setelah login berhasil, pengguna mendapat token yang dipakai sebagai autentikasi saat mengakses endpoint API yang dilindungi middleware.

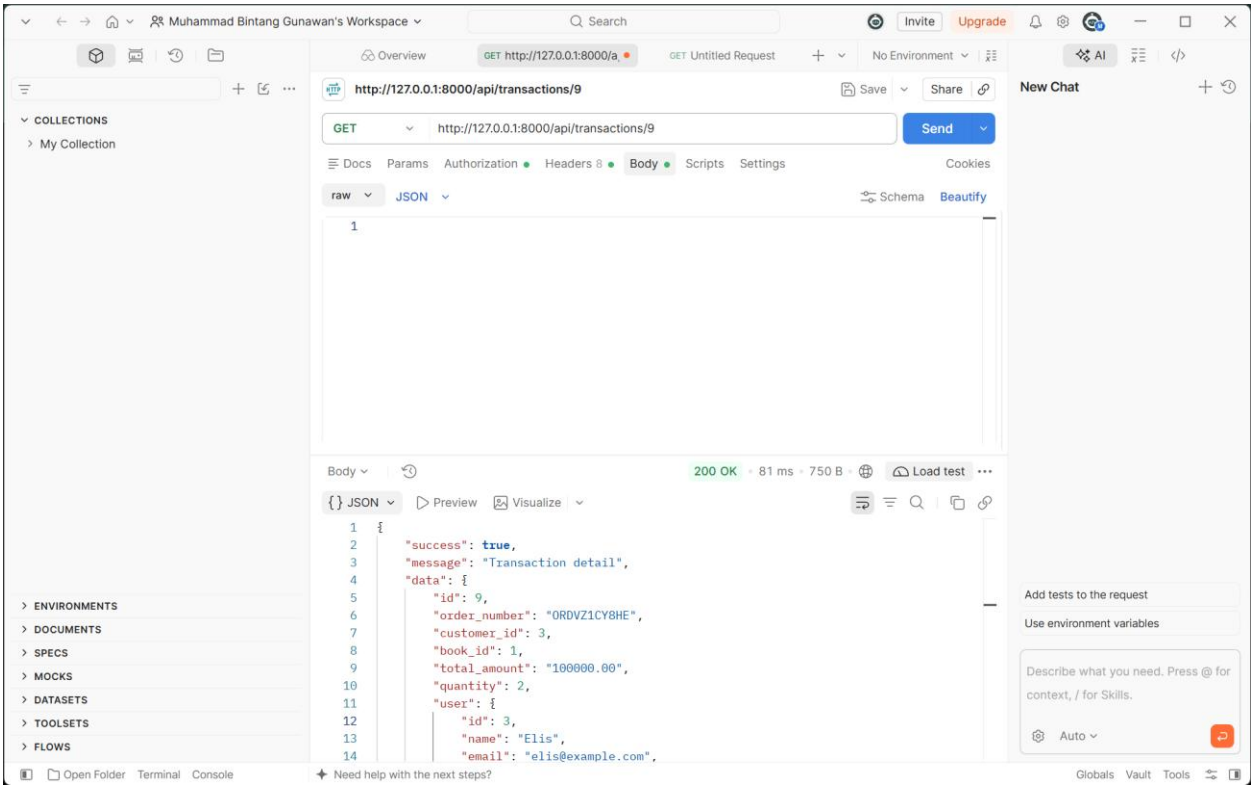
3. Hasil Pengujian/Screenshot

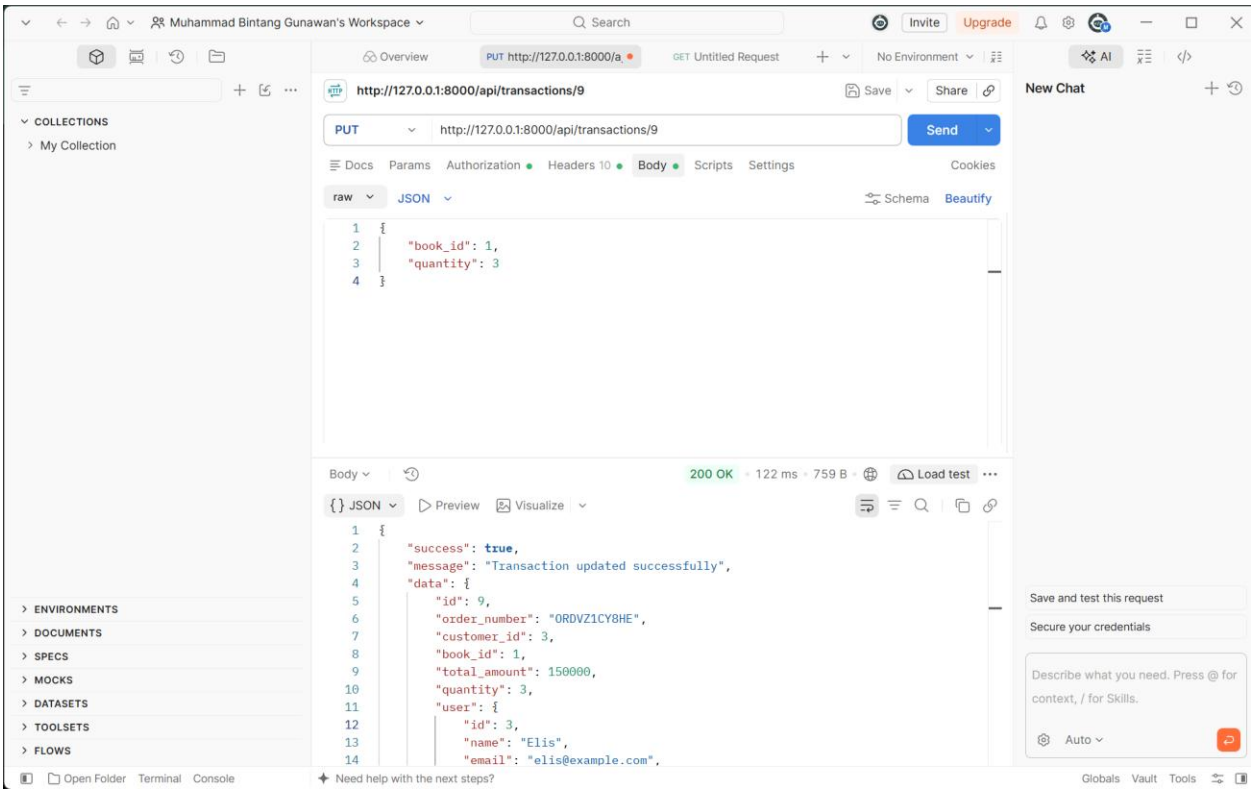


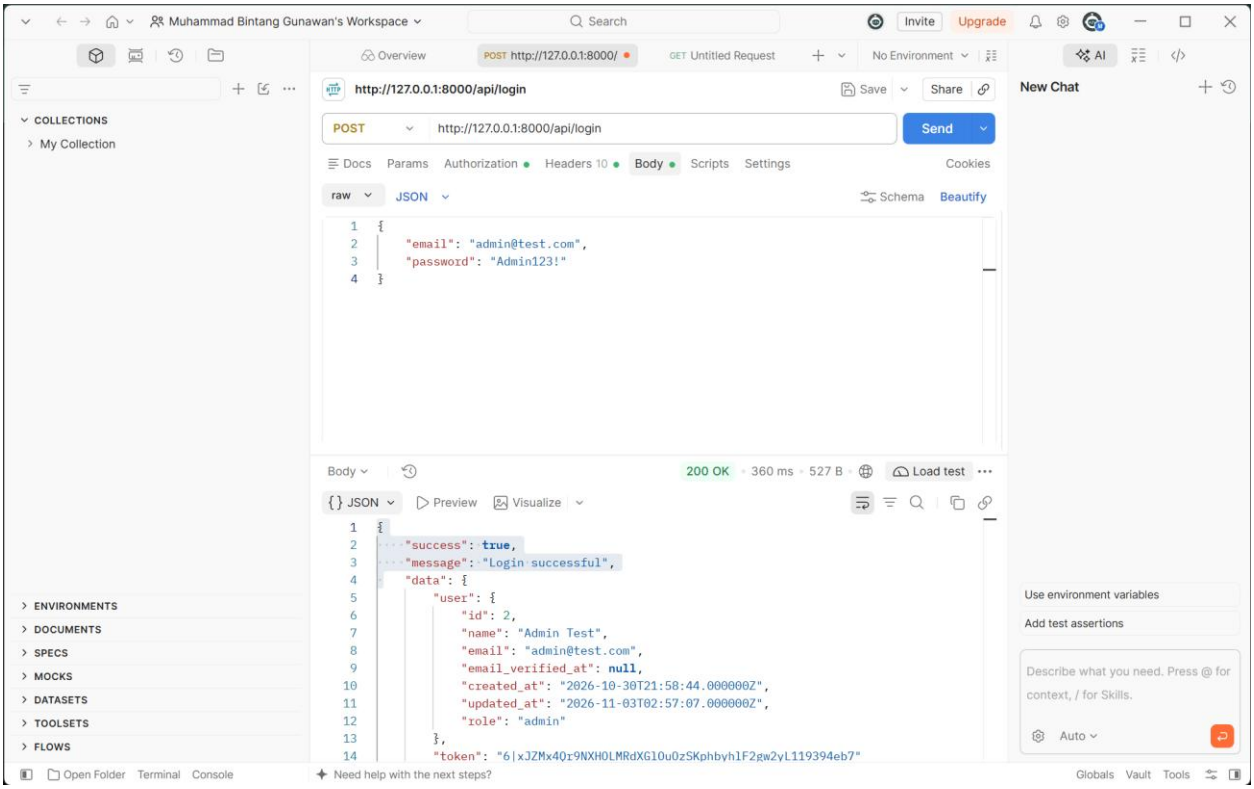


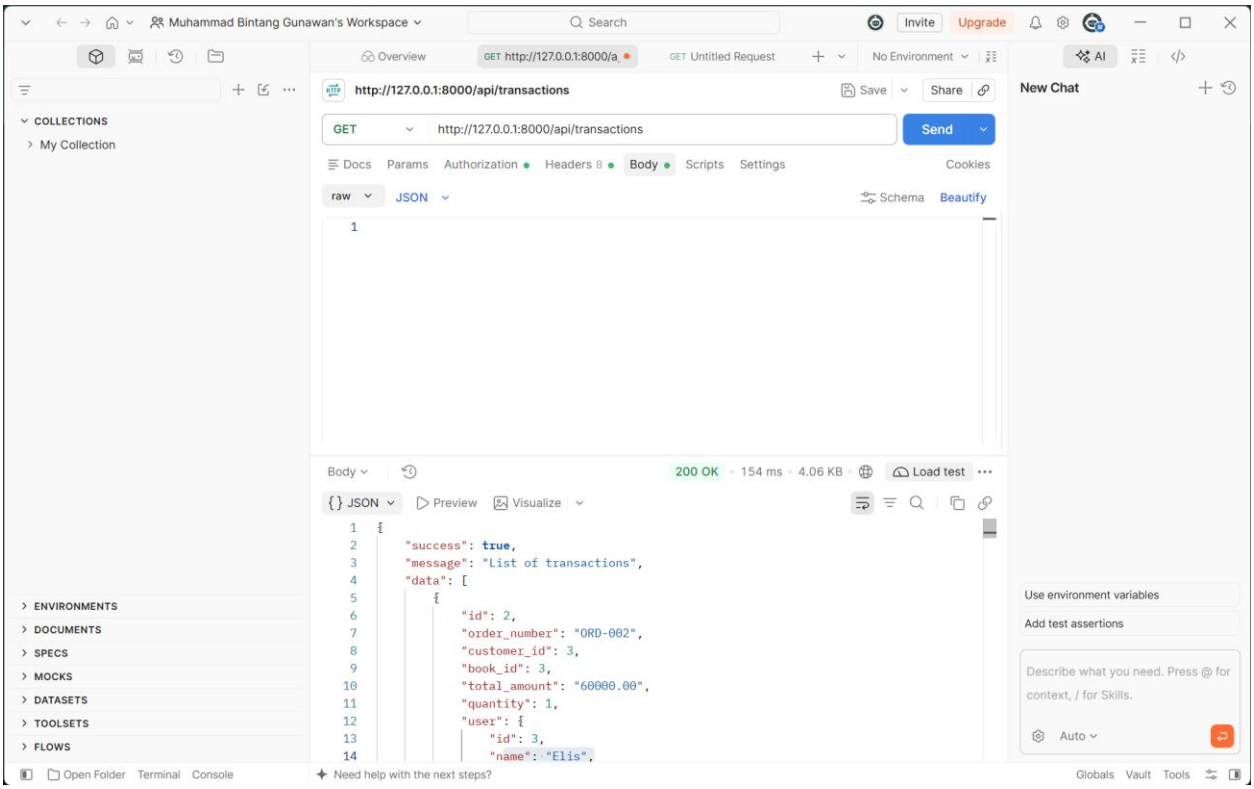

Muhammad Bintang Gunawan's Workspace
Search
Invite
Upgrade
Overview
POST http://127.0.0.1:8000/
GET Untitled Request
No Environment
AI
New Chat
COLLECTIONS
My Collection
http://127.0.0.1:8000/api/login
Save
Share
POST
http://127.0.0.1:8000/api/login
Send
Docs
Params
Authorization
Headers 10
Body
Scripts
Settings
Cookies
Auth type
Bearer Token
The authorization header will be automatically generated when you send the request. Learn more about authorization
Token
Body
Cookies
Headers 7
Test Results
200 OK
359 ms
522 B
Load test
JSON
Preview
Visualize
{
"success": true,
"message": "Login successful",
"data": {
"user": {
"id": 3,
"name": "Elis",
"email": "elis@example.com",
"email_verified_at": null,
"created_at": "2026-11-02T17:48:32.000000Z",
"updated_at": "2026-11-02T17:48:32.000000Z",
"role": "user"
},
"token": "5|svFCopoMHOUfAkIfpuevXGO2UGpgOdcXYnJklwxgba6145bf"
ENVIRONMENTS
DOCUMENTS
SPECS
MOCKS
DATASETS
TOOLSETS
FLOWS
Parameterize the auth token
Add response validation tests
Describe what you need. Press @ for context, / for Skills.
Auto
Open Folder
Terminal
Console
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Globals
Vault
Tools

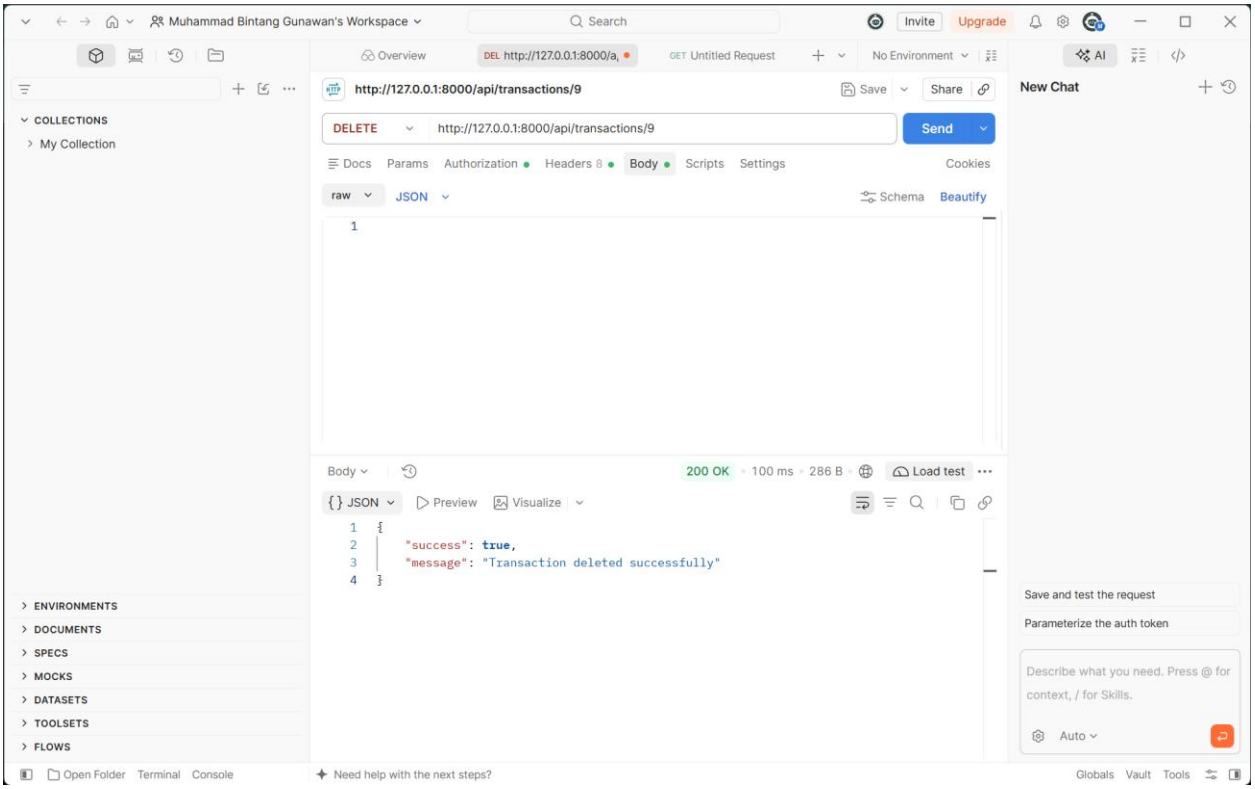

Muhammad Bintang Gunawan's Workspace
Search
Invite
Upgrade
Overview
POST http://127.0.0.1:8000/
GET Untitled Request
No Environment
AI
New Chat
COLLECTIONS
My Collection
http://127.0.0.1:8000/api/transactions
Save
Share
POST
http://127.0.0.1:8000/api/transactions
Send
Docs
Params
Authorization
Headers 10
Body
Scripts
Settings
Cookies
raw
JSON
Schema
Beautify
{
"book_id": 1,
"quantity": 2
}
Body
200 OK
208 ms
394 B
Load test
JSON
Preview
Visualize
{
"success": true,
"message": "Transaction created successfully",
"data": {
"order_number": "ORDVZ1CY8HE",
"customer_id": 3,
"book_id": 1,
"quantity": 2,
"total_amount": 100000,
"id": 9
}
}
ENVIRONMENTS
DOCUMENTS
SPECS
MOCKS
DATASETS
TOOLSETS
FLOWS
Debug the 500 error
Save request to collection
Describe what you need. Press @ for context, / for Skills.
Auto
Open Folder
Terminal
Console
Need help with the next steps?
Globals
Vault
Tools


Muhammad Bintang Gunawan's Workspace
Search
Invite
Upgrade
Overview
GET http://127.0.0.1:8000/a
GET Untitled Request
No Environment
AI
http://127.0.0.1:8000/api/transactions/9
Save
Share
New Chat
COLLECTIONS
My Collection
GET
http://127.0.0.1:8000/api/transactions/9
Send
Docs
Params
Authorization
Headers 8
Body
Scripts
Settings
Cookies
raw
JSON
Schema
Beautify
1
Body
200 OK
81 ms
750 B
Load test
JSON
Preview
Visualize
{
"success": true,
"message": "Transaction detail",
"data": {
"id": 9,
"order_number": "ORDVZ1CY8HE",
"customer_id": 3,
"book_id": 1,
"total_amount": "100000.00",
"quantity": 2,
"user": {
"id": 3,
"name": "Elis",
"email": "elis@example.com",
ENVIRONMENTS
DOCUMENTS
SPECS
MOCKS
DATASETS
TOOLSETS
FLOWS
Add tests to the request
Use environment variables
Describe what you need. Press @ for context, / for Skills.
Auto
Open Folder
Terminal
Console
Need help with the next steps?
Globals
Vault
Tools


Muhammad Bintang Gunawan's Workspace
Search
Invite
Upgrade
Overview
PUT http://127.0.0.1:8000/a
GET Untitled Request
No Environment
AI
http://127.0.0.1:8000/api/transactions/9
Save
Share
New Chat
COLLECTIONS
My Collection
PUT
http://127.0.0.1:8000/api/transactions/9
Send
Docs
Params
Authorization
Headers 10
Body
Scripts
Settings
Cookies
raw
JSON
Schema
Beautify
{
"book_id": 1,
"quantity": 3
}
Body
200 OK
122 ms
759 B
Load test
JSON
Preview
Visualize
{
"success": true,
"message": "Transaction updated successfully",
"data": {
"id": 9,
"order_number": "ORDVZ1CY8HE",
"customer_id": 3,
"book_id": 1,
"total_amount": 150000,
"quantity": 3,
"user": {
"id": 3,
"name": "Elis",
"email": "elis@example.com",
ENVIRONMENTS
DOCUMENTS
SPECS
MOCKS
DATASETS
TOOLSETS
FLOWS
Save and test this request
Secure your credentials
Describe what you need. Press @ for context, / for Skills.
Auto
Open Folder
Terminal
Console
Need help with the next steps?
Globals
Vault
Tools


Muhammad Bintang Gunawan's Workspace
Search
Invite
Upgrade
Overview
POST http://127.0.0.1:8000/
GET Untitled Request
No Environment
AI
New Chat
http://127.0.0.1:8000/api/login
Save
Share
POST
http://127.0.0.1:8000/api/login
Send
Docs
Params
Authorization
Headers 10
Body
Scripts
Settings
Cookies
raw
JSON
Schema
Beautify
{
"email": "admin@test.com",
"password": "Admin123!"
}
Body
200 OK
360 ms
527 B
Load test
JSON
Preview
Visualize
{
"success": true,
"message": "Login successful",
"data": {
"user": {
"id": 2,
"name": "Admin Test",
"email": "admin@test.com",
"email_verified_at": null,
"created_at": "2026-10-30T21:58:44.000000Z",
"updated_at": "2026-11-03T02:57:07.000000Z",
"role": "admin"
},
"token": "6|xJZMx4Qr9NXHOLMRdXGlOuOzSKphbvhlF2gw2yL119394eb7"
COLLECTIONS
My Collection
ENVIRONMENTS
DOCUMENTS
SPECS
MOCKS
DATASETS
TOOLSETS
FLOWS
Use environment variables
Add test assertions
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Auto
Open Folder
Terminal
Console
Need help with the next steps?
Globals
Vault
Tools


Muhammad Bintang Gunawan's Workspace
Search
Invite
Upgrade
Overview
GET http://127.0.0.1:8000/a
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AI
New Chat
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Save
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http://127.0.0.1:8000/api/transactions
Send
Docs
Params
Authorization
Headers 8
Body
Scripts
Settings
Cookies
raw
JSON
Schema
Beautify
Body
200 OK
154 ms
4.06 KB
Load test
JSON
Preview
Visualize
{
"success": true,
"message": "List of transactions",
"data": [
{
"id": 2,
"order_number": "ORD-002",
"customer_id": 3,
"book_id": 3,
"total_amount": "60000.00",
"quantity": 1,
"user": {
"id": 3,
"name": "Elis",
COLLECTIONS
My Collection
ENVIRONMENTS
DOCUMENTS
SPECS
MOCKS
DATASETS
TOOLSETS
FLOWS
Use environment variables
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Auto
Open Folder
Terminal
Console
Need help with the next steps?
Globals
Vault
Tools


Muhammad Bintang Gunawan's Workspace
Search
Invite
Upgrade
Overview
DEL http://127.0.0.1:8000/a,
GET Untitled Request
No Environment
AI
New Chat
http://127.0.0.1:8000/api/transactions/9
Save
Share
COLLECTIONS
My Collection
DELETE
http://127.0.0.1:8000/api/transactions/9
Send
Docs
Params
Authorization
Headers 8
Body
Scripts
Settings
Cookies
raw
JSON
Schema
Beautify
1
Body
200 OK
100 ms
286 B
Load test
{ } JSON
Preview
Visualize
1 {
2 "success": true,
3 "message": "Transaction deleted successfully"
4 }
ENVIRONMENTS
DOCUMENTS
SPECS
MOCKS
DATASETS
TOOLSETS
FLOWS
Save and test the request
Parameterize the auth token
Describe what you need. Press @ for context, / for Skills.
Auto
Open Folder
Terminal
Console
Need help with the next steps?
Globals
Vault
Tools